

Kyle Webb

Staff Data Scientist - statistics, data engineering, and applied AI
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PROFESSIONAL SUMMARY

Data scientist and data engineer with over ten years of experience spanning scientific research, federal health programs, and industry. Operates in both worlds: architects the data platforms, pipelines, and governance that make data trustworthy and readily usable, then applies machine learning and applied statistics to turn that data into predictions and decisions. Having built both sides, he knows how to serve data efficiently for modeling and predictive analytics - from raw ingestion to deployed model. Deep hands-on expertise across Databricks, AWS, and Terraform, paired with AI-augmented analytics and a statistician's rigor.

EXPERIENCE

eSimplicity - Staff Data Scientist

Sept 2023 - Present · Remote

Senior Data Scientist, Apr 2022 - Aug 2023 · AI Data Analyst Fellow, Rapid Innovation Center

- Led six DevOps and data engineers shipping validated data distributions across 20 CDR sources - terabytes of claims, provider, and beneficiary data orchestrated with AWS MWAA, Terraform, and Databricks.
- Spearheaded migration of all legacy EMR workloads to Databricks, building automated validation, testing, and monitoring - slashing median runtime 82% and median cost 93%.
- Architected data-governance workflows enforcing strict role-based access controls for 2,000+ users across a Kinesis-driven data repository.
- As AI Data Analyst Fellow, built and piloted an AI-agent cost-analysis app for Databricks spend, backed by an MCP server for live natural-language Q&A over cost data.
- Launched Databricks Delta Shares for governed cross-workspace sharing - eliminating duplicate data copies and their cost.

GovernmentCIO - Data Scientist

July 2021 - Apr 2022 · Remote

- Single-handedly automated VA IT infrastructure reporting in R/Python as the team's sole developer; briefed major incident data to Operations Triage Group leadership and presented to 200+ attendees including the VA Chief Information Officer.
- Delivered end-to-end analysis - querying, cleaning, modeling - for 30+ briefs spanning invalid security certificates to telecom outages for the largest U.S. healthcare network, several on one-day turnarounds.

National Human Genome Research Institute (NHGRI) - Bioinformatics Scientist

Oct 2020 - July 2021 · Bethesda, MD

- Stood up the lab's N3C data-enclave workspace from scratch; secured data use agreements for COVID-19 EHR pipelines and coached teammates on PySpark/SparkR.
- Curated and analyzed EHR data from millions of patients powering a COVID-19 Phenotype-Wide Association Study (PheWAS).
- Harmonized N3C and All of Us enclave data for a participant comparison study, applying stratified sampling across race, gender, age, and ethnicity to isolate true COVID-19 effects.

National Institute of Allergy and Infectious Diseases (NIAID) - Data Science Fellow

June 2018 - Sept 2020 · Bethesda, MD

- Engineered ETL pipelines and parallelized HPC jobs over 4TB+ of Cerner data to classify rare diseases with machine learning.
- Rebuilt performance and network visualization for SIGNAL, a public genetic screening tool (R Shiny + D3.js) - shipping in four months a roadmap scoped for six.
- De-identified 32,000 clinical notes, then trained and benchmarked two NLP models for Human Phenotype Ontology assignment, visualizing results with hierarchical t-SNE renderings.
- Extended the open-source R package ihazr, adding semiparametric hazard regression estimation and visualization.
- Completed five cross-institute data-science rotations - biostatistics, immunology, bioinformatics, grants analytics, epidemiology (detailed on CV).

EDUCATION

M.S., Statistics - Virginia Tech (2015 - 2016)

B.S., Statistics - Wake Forest University (2011 - 2015), GPA 3.4/4.0

CERTIFICATIONS

AWS Certified Cloud Practitioner (2023) · AWS Certified Solutions Architect - Associate (2026) · Databricks Certified Data Engineering Associate (2026)

TECHNICAL SKILLS

Python · R · SQL · Databricks · AWS (S3, EMR, Glue, MWAA, Kinesis) · Microsoft Azure · Terraform · RStudio · Git/Linux · JavaScript, D3.js